

LLD Case Study 15: Distributed Lock Manager (Redlock & Fencing Tokens)

The Bridge: Moving from 1 Server to 10 Servers! In Chapter 00-B, we learned about `threading.Lock()` —the "single bathroom key."

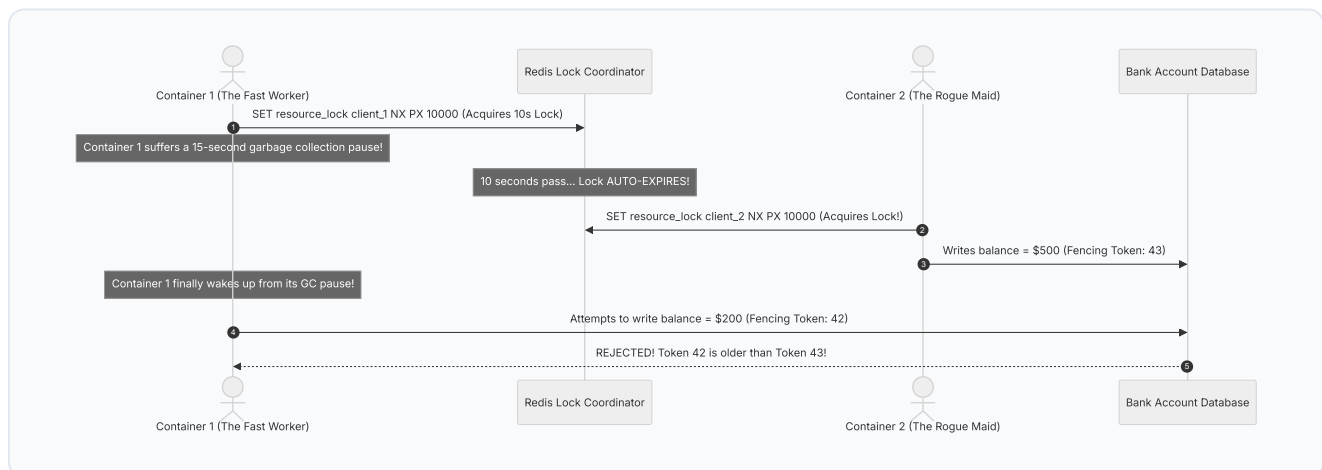
But what happens when your company scales, and your application runs on **10 separate Docker containers across AWS**?

A Python `threading.Lock` only protects memory inside a **single computer process**. Container 1 knows nothing about Container 2's memory! If Container 1 and Container 2 both attempt to charge customer Alice at the same second, both containers will succeed, charging her credit card twice!

To solve this, we need a **Distributed Lock Manager**—a shared external coordinator (like Redis) where all containers must acquire a digital keycard before touching critical data.

In this case study, you will build the industrial **Redlock Algorithm** and master **Fencing Tokens** (the #1 trap interviewers test at Staff level).

1. Zero-Prerequisite Intuition: The "Hotel Keycard & Rogue Maid"



The Problem: Why Simple Locks Break in Distributed Systems

1. **The Automatic Expiration (TTL):** In distributed systems, if a server acquires a lock and then crashes or gets hit by a lightning bolt, it can never release the lock. To prevent the entire company from deadlocking forever, all distributed locks have a **Time-To-Live (TTL)** (e.g. 10 seconds).

2. **The Danger (Long GC Pause):** What if Container 1 gets the lock, but then freezes for 15 seconds due to a full memory garbage collection pause?
 - While Container 1 is frozen, its 10-second lock expires!
 - Container 2 steps in, acquires the lock, and starts writing to the database.
 - Container 1 suddenly unfreezes, thinks it still holds the lock, and writes to the database at the exact same time as Container 2!
 3. **The Solution: Fencing Tokens (Monotonic Counters):**
 - Every time a lock is granted, the lock manager increments a counter: Token 42, Token 43, Token 44.
 - When writing to the database, the database **only accepts writes with a higher token number than the last write**.
 - Container 1's write (Token 42) is rejected because Token 43 already committed!
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2. Requirements & Architecture

Functional Requirements:

1. **Mutual Exclusion:** Only one client can hold the lock for a given resource at any instant.
 2. **Deadlock Freedom (Auto-Release):** Locks automatically expire after a configurable TTL if the holder crashes.
 3. **Fault Tolerance (Redlock):** Lock acquisition succeeds if a majority of independent Redis nodes ($N/2 + 1$) agree.
 4. **Fencing Token Generation:** Every successful acquisition returns a strictly increasing 64-bit integer token.
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3. Production-Grade Python Implementation



4. Milestone Check: What You Just Mastered!

You now understand:

1. Why local mutexes (`threading.Lock`) fail in multi-container cloud environments.
2. How the **Redlock Algorithm** reaches consensus across independent Redis nodes ($N/2 + 1$).

3. Why **Fencing Tokens** are mandatory to prevent split-brain write corruption during long Garbage Collection pauses.